UCL DEPARTMENT OF COMPUTER SCIENCE

COMP0080: Graphical Models

Comprehensive Past Exam Revision & Mock Exam Guide

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Course Core Topics Roadmap

This guide compiles all questions from past UCL COMP0080 examinations. The course covers three primary dimensions of probabilistic graphical models (PGMs) as detailed in David Barber's textbook *Bayesian Reasoning and Machine Learning* (BRML):

Topic	**Description & Key Algorithms Covered**
Representations	Directed models (Bayes Nets), Undirected models (Markov Nets), moralisation, triangulation
Exact Inference	Variable Elimination, Belief Propagation (Sum-Product), and Junction Tree Algorithm (JTA
Time Series	Hidden Markov Models (HMMs), forward filtering, backward smoothing, posterior sampling.
Learning	Maximum Likelihood (MLE) in fully observed networks, and EM algorithm for latent variabl
Approximate Inference	Variational Free Energy (mean-field approximations), VAEs, and MCMC / Gibbs / MH sam

PART 1: THE MOCK EXAM PAPER

Recommended Time: 3 Hours — Total Marks: 100
Instructions: Answer all questions. Show all intermediate work and justifications.

SECTION 1: Independence in Graphical Models [30 Marks]

Question 1.1: Independence in Joint Probability Tables (6 Marks)

Suppose that X_1 and X_2 are two random variables, where each can take one of three states. The joint distribution $P(X_1 = i, X_2 = j)$ is described by the probability table P:

$$P = \begin{pmatrix} 0.06 & 0.37 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.26 & 0.06 \end{pmatrix}$$

1. Are X_1 and X_2 independent? Support your answer. [2 Marks]

2. If they are not independent, what is the minimal number of entries in the table that you need to change to make them independent while keeping the table a valid joint probability distribution (non-negative and summing to 1.0)? State the changed entries and the new matrix. [4 Marks]

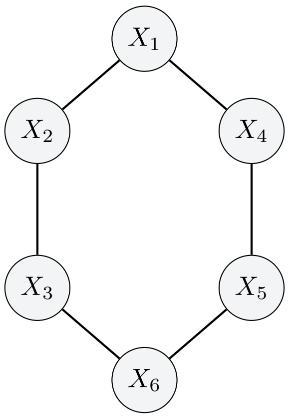
Question 1.2: Dependencies and Pairwise Independence (6 Marks)

1. Suppose that for three random variables A, B, and C, the following holds: $A \not\perp\!\!\!\perp B$ and $B \not\perp\!\!\!\perp C$. Does that necessarily imply that $A \not\perp\!\!\!\perp C$? Prove if it does, or provide a counterexample otherwise. [3 Marks]

2. Consider three random variables A, B, C. Give an example of a joint distribution in which $A \perp\!\!\!\perp B$, $B \perp\!\!\!\perp C$, $C \perp\!\!\!\perp A$ (pairwise independence), but A and B are **not** conditionally independent given C ($A \not\perp\!\!\!\perp B \mid C$). [3 Marks]

Question 1.3: Markov Network Marginal Independence (6 Marks)

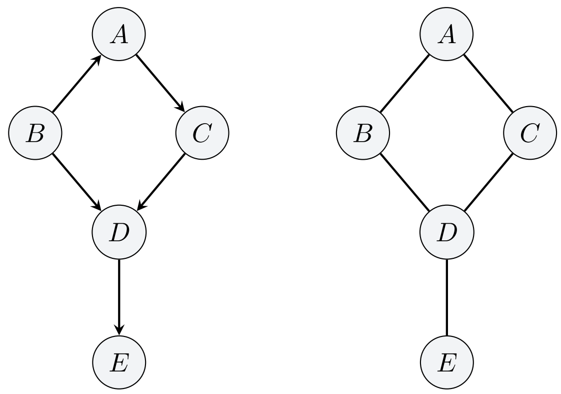
Consider the following Markov Network (undirected graph) on the variables $X_1, \ldots, X_6$:



Write down the set of all the conditional and marginal independence statements that are guaranteed to be true for the marginal distribution $P(X_2, X_4, X_6)$. Justify your answer using graph separation. [6 Marks]

Question 1.4: Markov Equivalence & Graph Comparisons (12 Marks)

1. Consider the following two graphical models (Model 1 and Model 2) on nodes A, B, C, D, E:



Model 1 (DAG) **Model 2 (Markov Network)**

Do they encode the same set of independence assumptions? If yes, explain why. If not, list all the independence statements that are true in one of the models, but not in the other. [6 Marks]

2. Consider the following four graphical models on three variables A, B, C:

   - **G1**: $A \to C \leftarrow B$ (Collider)
   - **G2**: $A \leftarrow C \to B$ (Common Cause)
   - **G3**: $A \to C \to B$ (Chain)
   - **G4**: $A \leftarrow C \leftarrow B$ (Reverse Chain)

   List all the independence statements that are encoded by each of these models. Do any of these encode the same set of independence statements? [6 Marks]

SECTION 2: Exact Inference and Parameter Learning [50 Marks]

Question 2.1: Bipartite Disease-Symptom Markov Network (15 Marks)

Consider a model of diseases and symptoms. $s_i \in \{0, 1\}$ is a binary random variable indicating whether the patient is showing the i-th symptom ($i = 1, \ldots, S$) and $d_j \in \{0, 1\}$ is a binary random variable indicating whether the patient has the j-th disease ($j = 1, \ldots, D$). A joint distribution for this is given by:

$$p(s, d) = \frac{1}{Z} \exp(s^T W d + a^T s + b^T d)$$

where Z is the normalization constant, and W, a, b are the parameters of the model.

1. Draw a Markov Network for this model. Describe its structure. [3 Marks]
2. Derive the expression for the conditional distribution $p(s \mid d)$. Is this distribution factorized? [3 Marks]
3. Suppose you have a dataset of N i.i.d. patient records (s^n, d^n). Derive the expression for the log-likelihood L of the parameters. [3 Marks]
4. Derive the expressions for the gradients $\frac{\partial L}{\partial W_{ij}}$, $\frac{\partial L}{\partial a_i}$, and $\frac{\partial L}{\partial b_j}$. [4 Marks]
5. Could these derivatives be computed efficiently in the general case? Justify your answer. [2 Marks]

Question 2.2: The Junction Tree Algorithm (15 Marks)

Consider a distribution that factorizes according to the DAG representing:

$$p(a, b, c, d, e, f) = p(a)p(b \mid a)p(c \mid b)p(d \mid c)p(e \mid d)p(f \mid a, e)$$

1. Draw the DAG representing this distribution. [3 Marks]
2. Draw the moralised version of this graph. Explain the moralisation step. [3 Marks]
3. Draw the triangulated version of the graph with the smallest possible maximum clique size. Explain your choice of chords. [3 Marks]
4. Draw a Junction Tree for this graph and verify that it satisfies the Running Intersection Property (RIP). [3 Marks]
5. Describe the initialization of clique potentials, the absorption procedure, and the associated message-passing schedule. [3 Marks]

Question 2.3: Inference & sampling in Hidden Markov Models (10 Marks)

Consider a Hidden Markov Model (HMM) with hidden states $h_{1:T} = \{h_1, \ldots, h_T\}$ and observed states $v_{1:T} = \{v_1, \ldots, v_T\}$.

1. When the sequence of outcomes $v_{1:T}$ is observed, it induces a posterior distribution on the hidden states $p_v(h_{1:T}) = p(h_{1:T} \mid v_{1:T})$. What is the graphical model of this posterior distribution? Based on this model, is h_1 independent of h_T? [3 Marks]
2. Suppose you want to sample from $p_v(h_{1:T})$. Is it possible to efficiently sample from it exactly? Describe how you would do it. [3 Marks]
3. Let $v_t \in \{0, 1, 2\}$, $h_t \in \{0, 1\}$. The parameters are:
   - Initial: $p(h_1 = 1) = 0.5$
   - Transition matrix: $T = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix}$ where $T_{ij} = p(h_t = i \mid h_{t-1} = j)$
   - Emission matrix: $B = \begin{pmatrix} 0.3 & 0.5 \\ 0.6 & 0 \\ 0.1 & 0.5 \end{pmatrix}$ where $B_{k,i} = p(v_t = k \mid h_t = i)$

   Suppose that you observe the sequence $v_{1:10} = [0, 1, 0, 2, 0, 2, 1, 0, 2, 0]$.

   - Is h_1 independent of h_{10} in $p_v(h_{1:10})$ given this specific sequence? Why? [2 Marks]
   - Compute the filtering distributions $p(h_8 \mid v_{1:8})$ and $p(h_9 \mid v_{1:9})$. [2 Marks]

Question 2.4: EM Parameter Estimation in Surveys (10 Marks)

To allow respondents to answer sensitive questions truthfully while maintaining confidentiality, a randomized response procedure is used. The respondent secretly tosses a biased coin (Coin 1) with heads probability p_1. If heads, they answer truthfully. If tails, they secretly toss a fair coin (Coin 2) and respond "Yes" for heads and "No" for tails. Let p_2 be the true fraction of the population that would answer "Yes" honestly.

1. Draw the graphical model corresponding to this scenario and specify CPTs. [2 Marks]
2. Suppose $p_1 = 0.5$. If we get a "Yes" response, what is the posterior probability that the person answered truthfully, as a function of p_2? [2 Marks]
3. Suppose $p_1 = 0.5$ and we observe a dataset D of responses. Write the likelihood and derive the MLE for p_2 (incorporating boundary constraints). [2 Marks]
4. Suppose both p_1 and p_2 are unknown and we only observe the dataset D. Derive the Expectation-Maximization (EM) updates for estimating p_1 and p_2. [2 Marks]
5. Let the dataset be $D = (\text{yes}, \text{yes}, \text{yes}, \text{no})$. Is the MLE unique when both p_1 and p_2 are unknown? Support your answer. [2 Marks]

SECTION 3: Approximate Inference and Sampling [20 Marks]

Question 3.1: KL Divergence Computation (5 Marks)

1. Explain what KL divergence is and define it mathematically. [2 Marks]
2. Suppose $X \in \{1, 2, 3, 4\}$ and $Y \in \{1, 2, 3, 4\}$. We have distributions $P(X) = (0.25, 0.25, 0.25, 0.25)^T$ and $Q(X) = (0.5, 0.25, 0.125, 0.125)^T$, and conditional matrices:

$$P(Y|X) = \begin{pmatrix} 0.25 & 0.5 & 0 & 0.2 \\ 0.25 & 0 & 0.5 & 0.3 \\ 0.25 & 0 & 0 & 0.1 \\ 0.25 & 0.5 & 0.5 & 0.4 \end{pmatrix}, \quad Q(Y|X) = \begin{pmatrix} 0.1 & 0 & 0 & 1 \\ 0 & 0 & 0.5 & 0 \\ 0.9 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \end{pmatrix}$$

   Compute the marginal distributions $P(Y)$ and $Q(Y)$, and calculate the divergences $KL(P(Y) \parallel Q(Y))$ and $KL(Q(Y) \parallel P(Y))$ to 4 decimal places. [3 Marks]

Question 3.2: Variational Free Energy Derivation (5 Marks)

Consider a pairwise Markov Network on three binary variables $X_1, X_2, X_3 \in \{0, 1\}$ with joint distribution:

$$P(X) = \frac{1}{Z}\Phi(X_1, X_2)\Phi(X_2, X_3)\Phi(X_3, X_1)$$

where $\Phi(X_i, X_j) = 2$ if $X_i = X_j$ and 1 if $X_i \neq X_j$. Derive the explicit expression for the variational free energy $F(q)$ using a fully factorized variational distribution $q(X) = \prod_{i=1}^{3} q_i(X_i)$ where $q_i(X_i = 1) = \mu_i$. Your expression must be explicit enough to be directly implemented in code. [5 Marks]

Question 3.3: Rejection Sampling on Lattices (5 Marks)

Consider a Markov Network on an $M \times N$ rectangular lattice where each $X_i \in \{0, 1\}$ and $P(X) = \frac{1}{Z}\prod_{i\sim j} e^{-\mathbb{I}(X_i \neq X_j)}$. Suppose we use rejection sampling with a uniform proposal distribution $Q(X) = 2^{-MN}$ to sample from $P(X)$.

1. Explain the rejection sampling procedure and define its average acceptance rate. [1 Mark]
2. Compute the exact average acceptance rate for:
   - $M = 2, N = 1$ [2 Marks]
   - $M = 2, N = 2$ [2 Marks]
3. Discuss whether rejection sampling is a good idea for larger lattice models. [1 Mark]

Question 3.4: MCMC, Gibbs, and Metropolis-Hastings (5 Marks)

1. Explain MCMC sampling and state the detailed balance condition. Is detailed balance a necessary condition for a sampler? Prove your answer by constructing a valid ergodic Markov chain sampler that does not satisfy detailed balance. [3 Marks]
2. Prove that Gibbs sampling satisfies detailed balance. [1 Mark]
3. Prove that the Metropolis-Hastings algorithm satisfies detailed balance. [1 Mark]

PART 2: EXHAUSTIVE SOLUTIONS GUIDE

SECTION 1: Independence in Graphical Models

Solution 1.1: Independence in Joint Probability Tables

1. **Testing Independence:** For independence, we require $P_{ij} = p(X_1 = i)p(X_2 = j) = r_i \cdot c_j$, where r_i is the i-th row marginal sum, and c_j is the j-th column marginal sum. Let's compute the marginals:

   - Row marginals (sums across columns):

   $$r_1 = 0.06 + 0.37 + 0.12 = 0.55$$

   $$r_2 = 0.01 + 0.07 + 0.02 = 0.10$$

   $$r_3 = 0.03 + 0.26 + 0.06 = 0.35$$

   - Column marginals (sums across rows):

   $$c_1 = 0.06 + 0.01 + 0.03 = 0.10$$

   $$c_2 = 0.37 + 0.07 + 0.26 = 0.70$$

   $$c_3 = 0.12 + 0.02 + 0.06 = 0.20$$

   Now we check if $P_{11} = r_1 \cdot c_1$:

   $$r_1 \cdot c_1 = 0.55 \times 0.10 = 0.055 \neq 0.06 = P_{11}$$

   Since $P_{11} \neq r_1 c_1$, the variables X_1 and X_2 are **not independent**.

2. **Minimal Changes to Make Independent:** An independent joint probability table represents a rank-1 matrix ($P = uv^T$). Let's inspect the rows: - Row 2 is $R_2 = [0.01, 0.07, 0.02]$. - Check the ratio of columns 1 and 3 across all rows: - Row 1: $[0.06, 0.12] = 6 \times [0.01, 0.02]$ - Row 2: $[0.01, 0.02] = 1 \times [0.01, 0.02]$ - Row 3: $[0.03, 0.06] = 3 \times [0.01, 0.02]$ Columns 1 and 3 are already in a perfect ratio of $6 : 1 : 3$. For the entire matrix to be rank-1, column 2 must follow this same ratio. Setting $P_{22} = 0.07$ as our reference: - $P_{12} = 6 \times P_{22} = 0.42$ (currently 0.37) - $P_{32} = 3 \times P_{22} = 0.21$ (currently 0.26)

   If we change exactly **2 entries** ($P_{12} \leftarrow 0.42$ and $P_{32} \leftarrow 0.21$), the new matrix is:

   $$P_{\text{new}} = \begin{pmatrix} 0.06 & 0.42 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.21 & 0.06 \end{pmatrix}$$

   This matrix is rank-1, sums to exactly 1.0 ($0.60 + 0.10 + 0.30 = 1.00$), and contains only non-negative entries. Since we cannot satisfy the rank-1 constraint on both rows simultaneously by changing only 1 entry, the minimal number of entries to change is **2**.

Solution 1.2: Dependencies and Pairwise Independence

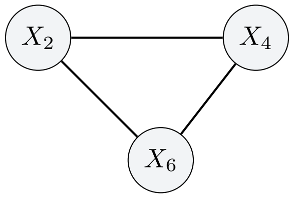
1. **Proof/Counterexample for transitivity:** No, dependency is not transitive. *Counterexample:* Let A and C be two independent fair coin tosses ($A \perp\!\!\!\perp C$). Let $B = (A, C)$ be the joint variable containing both outcomes. Since B contains A, A and B are dependent ($A \not\perp\!\!\!\perp B$). Since B contains C, B and C are dependent ($B \not\perp\!\!\!\perp C$). Yet, $A \perp\!\!\!\perp C$ by construction.

2. **Pairwise Independence but Conditional Dependence (XOR):** Let $A, B \in \{0, 1\}$ be independent fair coins ($P(A = a) = 0.5$, $P(B = b) = 0.5$). Let $C = A \oplus B$ (XOR). - $P(A, B) = P(A)P(B) = 0.25$ by construction. - $P(B = b, C = c) = P(B = b, A \oplus B = c) = P(B = b, A = b \oplus c) = 0.25 = P(B)P(C)$ since $P(C = c) = 0.5$. Hence $B \perp\!\!\!\perp C$. - Similarly, $C \perp\!\!\!\perp A$. - However, given $C = 0$, $A \oplus B = 0 \implies A = B$. Thus, $P(A = 1, B = 1 \mid C = 0) = 0.5 \neq 0.25 = P(A = 1 \mid C = 0)P(B = 1 \mid C = 0)$. They are conditionally dependent given C.

Solution 1.3: Markov Network Marginal Independence

The target is the marginal distribution $P(X_2, X_4, X_6)$, where X_1, X_3, X_5 are marginalized out. In a Markov Network, marginalization removes nodes and creates cliques (fully connected subgraphs) between all their immediate neighbors: - Marginalizing out X_1 (connected to X_2 and X_4) connects X_2 and X_4 directly. - Marginalizing out X_3 (connected to X_2 and X_6) connects X_2 and X_6 directly. - Marginalizing out X_5 (connected to X_4 and X_6) connects X_4 and X_6 directly.

As a result, the marginal distribution $P(X_2, X_4, X_6)$ factorizes over a fully connected triangle:



Since the marginalized graph is a clique, no separation statements hold. The set of guaranteed independence statements is the **empty set ($\emptyset$)**.

Solution 1.4: Markov Equivalence & Graph Comparisons

1. **Model 1 vs Model 2: - Skeleton:** Both have the same skeleton: $\{(A, B), (A, C), (B, D), (C, D), (D, E)\}$. - **Immoralities:** - Model 1 (DAG) has one immorality: $B \to D \leftarrow C$ (collider at D with unconnected parents B and C). - Model 2 (Markov Network) is undirected and has no arrowheads, hence no colliders and no immoralities.

   Since they have different immoralities, they do not encode the same set of independence assumptions. - In Model 1 (DAG): $B \perp\!\!\!\perp C \mid A$ is true (blocked path through A). In Model 2 (Markov Net), $B \perp\!\!\!\perp C \mid A$ is false (path $B - D - C$ remains open). - In Model 1 (DAG): $B \not\!\perp\!\!\!\perp C \mid \{A, D\}$ is true (collider D is open). In Model 2 (Markov Net), $B \perp\!\!\!\perp C \mid \{A, D\}$ is true (blocked by A and D).

2. **Comparing Three-Node DAGs: - G1** (Collider) encodes $\{A \perp\!\!\!\perp B\}$. - **G2** (Common Cause), **G3** (Chain), **G4** (Reverse Chain) all encode the same statement: $\{A \perp\!\!\!\perp B \mid C\}$. They are Markov equivalent.

SECTION 2: Exact Inference and Parameter Learning

Solution 2.1: Bipartite Disease-Symptom Markov Network

1. **Markov Network Structure:** The energy function contains only single-variable terms ($a^T s$, $b^T d$) and cross-terms ($s^T W d = \sum_{i,j} W_{ij} s_i d_j$). There are no symptom-symptom or disease-disease connections. Thus, the Markov Network is a **bipartite graph** with symptoms s_i on one side and diseases d_j on the other. An edge exists between s_i and d_j if and only if $W_{ij} \neq 0$.

2. **Conditional Distribution $p(s \mid d)$:**

$$p(s \mid d) = \frac{\exp(s^T W d + a^T s)}{\sum_{s'} \exp(s'^T W d + a^T s')} = \frac{\prod_i \exp(s_i(W_i.d + a_i))}{\prod_i (1 + \exp(W_i.d + a_i))} = \prod_{i=1}^{S} \frac{\exp(s_i(W_i.d + a_i))}{1 + \exp(W_i.d + a_i)}$$

   This completely factorizes as $p(s \mid d) = \prod_{i=1}^{S} p(s_i \mid d)$, where $p(s_i = 1 \mid d) = \sigma(W_i.d + a_i)$.

3. **Log-Likelihood:**

$$L = \sum_{n=1}^{N} \left[(s^n)^T W d^n + a^T s^n + b^T d^n\right] - N \log Z(W, a, b)$$

   where $Z(W, a, b) = \sum_{s,d} \exp(s^T W d + a^T s + b^T d)$.

4. **Gradients:**

$$\frac{\partial L}{\partial W_{ij}} = \sum_{n=1}^{N} s_i^n d_j^n - N \frac{\partial \log Z}{\partial W_{ij}} = \sum_{n=1}^{N} \left(s_i^n d_j^n - \langle s_i d_j \rangle_{p(s,d)}\right)$$

$$\frac{\partial L}{\partial a_i} = \sum_{n=1}^{N} \left(s_i^n - \langle s_i \rangle_{p(s,d)}\right), \quad \frac{\partial L}{\partial b_j} = \sum_{n=1}^{N} \left(d_j^n - \langle d_j \rangle_{p(s,d)}\right)$$

5. **Efficiency:** No. Evaluating the model expectations $\langle s_i d_j \rangle_{p(s,d)}$ requires summing over all 2^{S+D} states to compute Z. This is NP-hard for general graphs.

Solution 2.2: The Junction Tree Algorithm

1. **DAG Graph:** Directed edges: $a \to b$, $b \to c$, $c \to d$, $d \to e$, $a \to f$, $e \to f$.

2. **Moralisation:** Marry parents of f (a and e), then convert all edges to undirected. Moral edge is $a - e$.

3. **Triangulation:** The moralised graph has a cycle of length 5: $a - b - c - d - e - a$. To break it into triangles, we add chords $a - c$ and $a - d$. The resulting cliques are all of size 3 (minimal).

4. **Junction Tree and RIP:** Cliques: $C_1 = \{a, b, c\}$, $C_2 = \{a, c, d\}$, $C_3 = \{a, d, e\}$, $C_4 = \{a, e, f\}$. Connected as a chain:

$$\boxed{\{a,b,c\}} \overset{\{a,c\}}{\text{———}} \boxed{\{a,c,d\}} \overset{\{a,d\}}{\text{———}} \boxed{\{a,d,e\}} \overset{\{a,e\}}{\text{———}} \boxed{\{a,e,f\}}$$

   RIP is satisfied because any variable shared between two cliques lies on the unique path between them (e.g., $a \in C_1 \cap C_4$ is in C_2, C_3).

5. **JTA Steps:** - Potentials: $\psi(C_1) = p(a)p(b|a)p(c|b)$, $\psi(C_2) = p(d|c)$, $\psi(C_3) = p(e|d)$, $\psi(C_4) = p(f|a, e)$. Separators $\phi(S) = 1$. - Absorption from W to V via S: $\phi(S)^{\text{new}} = \sum_{W \setminus S} \psi(W)$; $\psi(V)^{\text{new}} = \psi(V)\frac{\phi(S)^{\text{new}}}{\phi(S)^{\text{old}}}$; $\phi(S)^{\text{old}} = \phi(S)^{\text{new}}$. - Schedule: Forward $C_1 \to C_2 \to C_3 \to C_4$, Backward $C_4 \to C_3 \to C_2 \to C_1$.

Solution 2.3: Inference & sampling in Hidden Markov Models

1. **Posterior Graph:** $h_1 \to h_2 \to \cdots \to h_T$. Since it's a directed path with no colliders, h_1 and h_T are not independent.

2. **Sampling:** Compute filtering messages $\alpha_t(h_t) = p(h_t, v_{1:t})$. Sample $h_T \sim \alpha_T(h_T)$. Then sample backwards: $h_t \sim p(h_{t+1} \mid h_t)\alpha_t(h_t)$.

3. **Numerical HMM: - Decoupling:** $B_{1,1} = p(v_t = 1 \mid h_t = 1) = 0$. Since $v_2 = 1$ and $v_7 = 1$, we have $p(h_2 = 0 \mid v) = 1$ and $p(h_7 = 0 \mid v) = 1$. Since h_2 is instantiated, it d-separates h_1 from h_{10}. They are conditionally independent. **- Filtering Distributions:** Start from $p(h_7 \mid v_{1:7}) = \begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix}$. - $p(h_8 \mid v_{1:7}) = Tp(h_7 \mid v_{1:7}) = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$. - For $v_8 = 0$, $p(h_8 \mid v_{1:8}) \propto \begin{pmatrix} 0.3 \times 0.5 \\ 0.5 \times 0.5 \end{pmatrix} = \begin{pmatrix} 0.15 \\ 0.25 \end{pmatrix} \implies \begin{pmatrix} \mathbf{0.375} \\ \mathbf{0.625} \end{pmatrix}$. - $p(h_9 \mid v_{1:8}) = Tp(h_8 \mid v_{1:8}) = \begin{pmatrix} 0.5 \times 0.375 + 0.8 \times 0.625 \\ 0.5 \times 0.375 + 0.2 \times 0.625 \end{pmatrix} = \begin{pmatrix} 0.6875 \\ 0.3125 \end{pmatrix}$. - For $v_9 = 2$, $p(h_9 \mid v_{1:9}) \propto \begin{pmatrix} 0.1 \times 0.6875 \\ 0.5 \times 0.3125 \end{pmatrix} = \begin{pmatrix} 0.06875 \\ 0.15625 \end{pmatrix} \implies \begin{pmatrix} \mathbf{11/36} \\ \mathbf{25/36} \end{pmatrix} \approx \begin{pmatrix} \mathbf{0.3056} \\ \mathbf{0.6944} \end{pmatrix}$.

Solution 2.4: EM Parameter Estimation in Surveys

1. **Graphical Model:** $H \to Y \leftarrow C_1$. CPT: $P(Y = 1 \mid H, C_1) = C_1 H + (1 - C_1) \cdot 0.5$.

2. **Posterior:** $P(C_1 = 1 \mid Y = 1) = \frac{p_2}{p_2 + 0.5} = \frac{2p_2}{2p_2 + 1}$.

3. **MLE:** $\theta = P(Y = 1) = 0.5p_2 + 0.25 \implies \hat{p}_2 = \max\left(0, \min\left(1, 2\frac{n_y}{n} - 0.5\right)\right)$.

4. **EM Updates:** - E-step: - If $Y_i = 1$: $\langle C_{1,i} \rangle = \frac{p_1 p_2}{p_1 p_2 + 0.5(1 - p_1)}$, $\langle H_i \rangle = \frac{p_2(p_1 + 0.5(1 - p_1))}{p_1 p_2 + 0.5(1 - p_1)}$. - If $Y_i = 0$: $\langle C_{1,i} \rangle = \frac{p_1(1 - p_2)}{p_1(1 - p_2) + 0.5(1 - p_1)}$, $\langle H_i \rangle = \frac{0.5(1 - p_1)p_2}{p_1(1 - p_2) + 0.5(1 - p_1)}$. - M-step: $p_1^{\text{new}} = \frac{1}{n}\sum_i \langle C_{1,i} \rangle$, $p_2^{\text{new}} = \frac{1}{n}\sum_i \langle H_i \rangle$.

5. **Uniqueness:** Not unique. For $n_y = 3, n = 4$, any pair satisfying $p_1(p_2 - 0.5) = 0.25$ is an MLE (e.g., $(0.5, 1.0)$ and $(1.0, 0.75)$).

SECTION 3: Approximate Inference and Sampling

Solution 3.1: KL Divergence Computation

1. **Definition:** $KL(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$.

2. **Marginals and KL:** - $P(Y = i) = \frac{1}{4} \sum_j P_{ij} \implies P(Y) = [0.2375, 0.2625, 0.0875, 0.4125]^T$ - $Q(Y = i) = \sum_j Q_{ij} Q(X_j) \implies Q(Y) = [0.1750, 0.0625, 0.5750, 0.1875]^T$ - $KL(P(Y) \parallel Q(Y)) = 0.2375 \log \frac{0.2375}{0.1750} + 0.2625 \log \frac{0.2625}{0.0625} + 0.0875 \log \frac{0.0875}{0.5750} + 0.4125 \log \frac{0.4125}{0.1875} = \mathbf{0.6097}$ nats (0.8797 bits). - $KL(Q(Y) \parallel P(Y)) = \mathbf{0.7916}$ nats (1.1420 bits).

Solution 3.2: Variational Free Energy Derivation

Entropy term $\langle \log q(X) \rangle_q = \sum_{i=1}^{3} [\mu_i \log \mu_i + (1 - \mu_i) \log(1 - \mu_i)]$.
Energy term $\langle \log \tilde{P}(X) \rangle_q = \log 2 \sum_{i \sim j} [\mu_i \mu_j + (1 - \mu_i)(1 - \mu_j)]$.
Free energy $F(q) = \sum_{i=1}^{3} [\mu_i \log \mu_i + (1 - \mu_i) \log(1 - \mu_i)] - \log 2 \sum_{i \sim j} [\mu_i \mu_j + (1 - \mu_i)(1 - \mu_j)]$, where the second sum is over edges $(1, 2)$, $(2, 3)$, and $(3, 1)$.

Solution 3.3: Rejection Sampling on Lattices

1. **Procedure:** Propose $X \sim Q(X) = 2^{-MN}$, accept with probability $a(X) = \tilde{P}(X)$. Average acceptance rate is $P(\text{accept}) = Z/2^{MN}$.

2. **Calculations:** - $M = 2, N = 1$ (2 nodes, 1 edge): $Z = 2 + 2e^{-1} \approx 2.7358 \implies P(\text{accept}) = \mathbf{0.6839}$ (68.4%). - $M = 2, N = 2$ (4 nodes, 4 edges): $Z = 2 + 12e^{-2} + 2e^{-4} \approx 3.6606 \implies P(\text{accept}) = \mathbf{0.2288}$ (22.9%).

3. **Scaling:** Poor. Acceptance rate scales as $Z/2^{MN}$, which decays exponentially to 0 as $MN \to \infty$.

Solution 3.4: MCMC, Gibbs, and Metropolis-Hastings

1. **MCMC Detailed Balance:** Detailed balance is sufficient, not necessary. *Counterexample:* Target $P = [1/3, 1/3, 1/3]^T$, lazy cyclic transition $T = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$. $PT = P$ (stationary), but $P(1)T(2 \mid 1) = 1/6 \neq 0 = P(2)T(1 \mid 2)$ (violates detailed balance).

2. **Gibbs Sampling:** Let $x = (x_i, x_{-i})$ and $x' = (x'_i, x_{-i})$.

$$P(x)T(x' \mid x) = p(x_i, x_{-i}) p(x'_i \mid x_{-i}) = p(x_{-i}) p(x_i \mid x_{-i}) p(x'_i \mid x_{-i})$$

$$P(x')T(x \mid x') = p(x'_i, x_{-i}) p(x_i \mid x_{-i}) = p(x_{-i}) p(x'_i \mid x_{-i}) p(x_i \mid x_{-i})$$

Since both sides are equal, detailed balance holds.

3. **Metropolis-Hastings:** For $x \neq x'$, $T(x' \mid x) = q(x' \mid x) a(x', x)$.

$$P(x)T(x' \mid x) = P(x) q(x' \mid x) \min\left(1, \frac{P(x') q(x \mid x')}{P(x) q(x' \mid x)}\right) = \min\left(P(x) q(x' \mid x), P(x') q(x \mid x')\right)$$

By symmetry, this is equal to $P(x')T(x \mid x')$. Detailed balance holds.